

# Star chromatic index of cubic graphs

|           |                |                      |              |
|-----------|----------------|----------------------|--------------|
| OPG-37271 | L1 - Tractable | Partiellement résolu | Graph Theory |
|-----------|----------------|----------------------|--------------|

## Énoncé original (anglais, reproduction mot pour mot)

The star chromatic index  $\chi'_s(G)$  of a graph  $G$  is the minimum number of colors needed to properly color the edges of the graph so that no path or cycle of length four is bi-colored.

Question Is it true that for every (sub)cubic graph  $G$ , we have  $\chi'_s(G) \leq 6$ ?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>